

EDS Practical :04
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PRACTICAL: 04

4.1.1. Pandas - series creation and manipulation

Write a Python program that takes a list of numbers from the user, creates a Pandas series from it, and then calculates the mean of even and odd numbers separately using the groupby and mean() operations.

Hint: You can group the Series based on whether each number is even or odd.

Input Format:

- The user should enter a list of numbers separated by space when prompted.

Output Format:

- The program should display the mean of even and odd numbers separately.
- Each mean value should be displayed with a label indicating whether it corresponds to even or odd numbers.

Refer to the sample test cases for better understanding regarding input and output format.

```
# import pandas as pd
```

```
## Take inputs from the user to create a list of numbers
```

```
# numbers = list(map(int, input().split()))
```

```
# grouped =
```

```
## Display the mean of even and odd numbers with labels
```

```
# grouped.index = ['Even' if is_even else 'Odd' for is_even in grouped.index]
```

```
# print("Mean of even and odd numbers:")
```

```
# print(grouped) import pandas as
```

```
pd
```

```
# Take inputs from the user to create a list of numbers numbers
```

```
= list(map(int, input().split()))
```

```
# Create a Pandas series from the list of numbers s
```

```
= pd.Series(numbers)
```

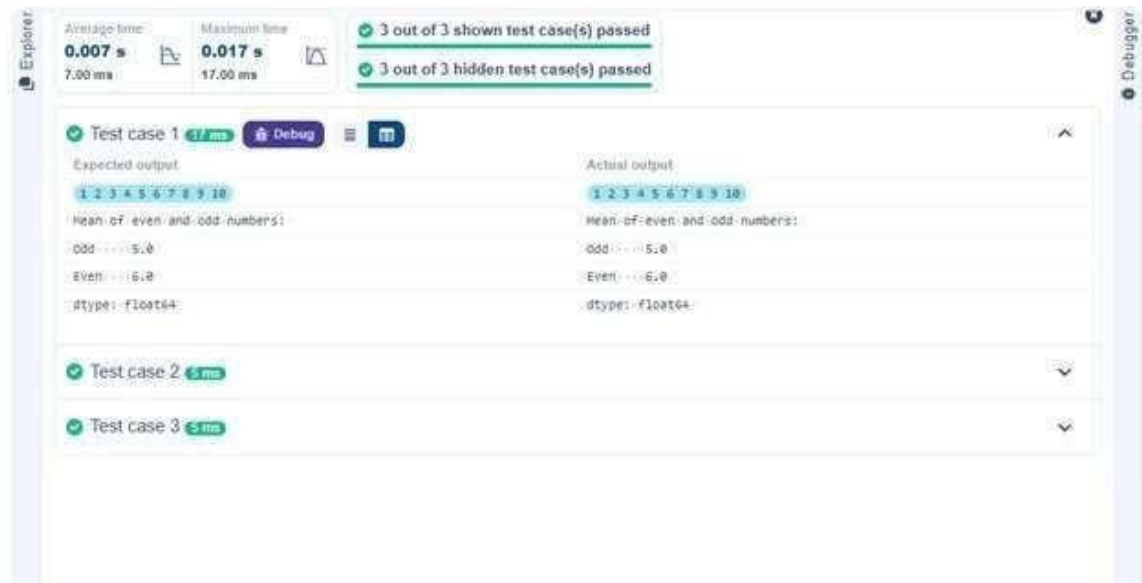
```
# Grouping by even and odd numbers and calculating the mean grouped
```

```
= s.groupby(s % 2 == 0).mean()
```

```
# Display the mean of even and odd numbers with labels grouped.index = ['Even'
```

```
if is_even else 'Odd' for is_even in grouped.index] print("Mean of even and
```

```
odd numbers:") print(grouped)
```



4.1.2. Dictionary to dataframe

07:26

A dictionary of lists has been provided to you in the editor. Create a DataFrame from the dictionary of lists and perform the listed operations, then display the DataFrame before and after each manipulation.

Create the DataFrame:

- Convert the dictionary to a Pandas DataFrame.

Add a new row:

- Take inputs from the user for the new row data (name, age).
- Add the new row to the DataFrame.
- Display the DataFrame after adding the new row.

Modify a row:

- Modify a specific row by changing the age. Take the row index and new age value from the user.
- Display the DataFrame after modifying the row.

Delete a row:

- Take the row index to be deleted from the user.
- Remove the specified row.
- Display the DataFrame after deleting the row.

Add a new column:

- Add a column Gender with values taken from the user.
- Display the DataFrame after adding the new column.

Modify a column:

- Convert names to uppercase.
- Display the DataFrame after modifying the column.

Delete a column:

- Remove the Age column.
- Display the DataFrame after deleting the column.

4.1.3. Student Information

01:49

Write a program to read a text file containing student information (name, age, and grade) using Pandas. Perform the following tasks:

- Display the first five rows of the data frame.
- Calculate the average age of the students (limit the average age up to 2 decimal places).
- Filter out the students who have a grade above a certain threshold (consider the threshold grade is 'B').

Note:

Refer to the displayed test cases for better understanding.

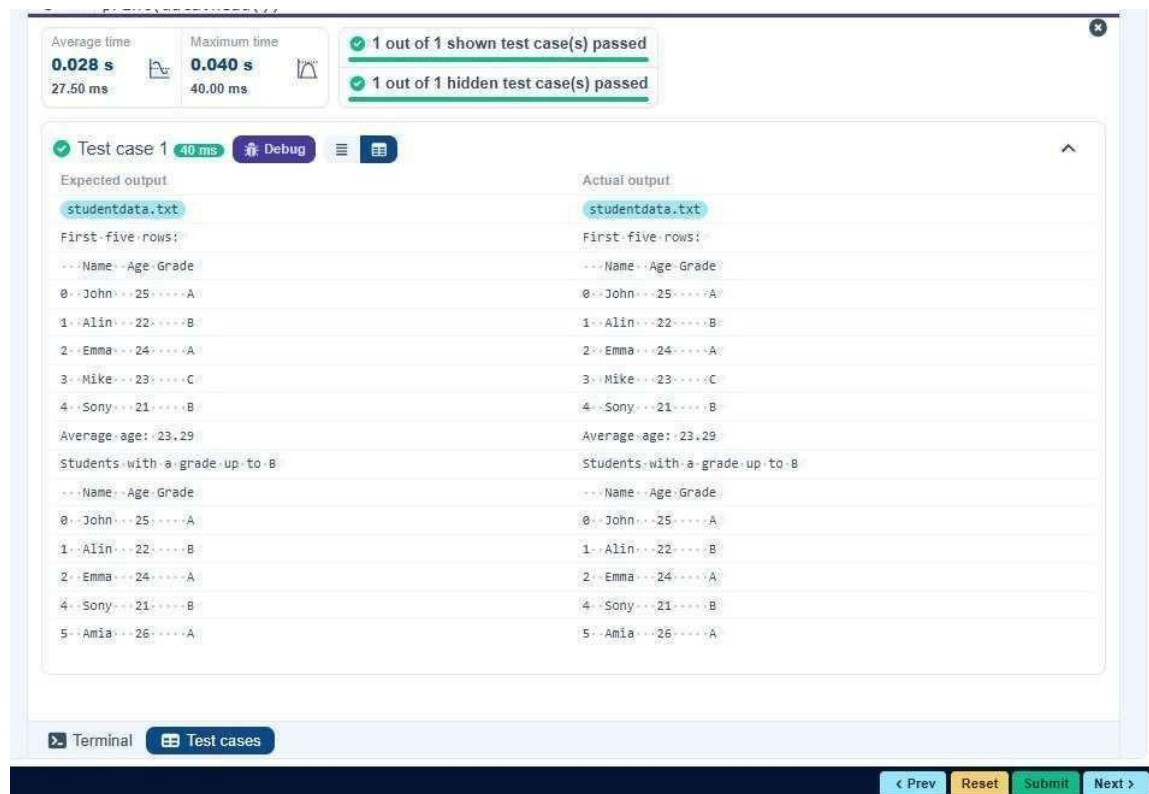
```
import pandas as pd
```

```
# Read the text file into a DataFrame file = input() data = pd.read_csv(file, sep="\s+",  
header=None, names=["Name", "Age", "Grade"])
```

```
# Display the first five rows print("First five  
rows:") print(data.head())
```

```
# Calculate the average age (2 decimal places) avg_age  
= round(data["Age"].mean(), 2) print("Average age:",  
avg_age)
```

```
# Filter students with grade up to 'B' (A and B) filtered  
= data[data["Grade"].isin(['A', 'B'])] print("Students  
with a grade up to B") print(filtered)
```



4.2.1. Month with the Highest Total Sales

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Group the data by Month and calculate the total sales for each month.
- Find the month with the highest total sales and display it.
- Also, display the total sales for the best month.

Sample Data:

Date,Product,Quantity,Price,City

2025-01-01,Product A,5,20,New York

2025-01-01,Product B,3,15,Los Angeles

2025-01-02,Product A,7,20,New York

2025-01-02,Product C,4,30,Chicago

2025-01-03,Product B,2,15,Chicago

2025-01-03,Product A,8,20,Los Angeles

2025-01-04,Product C,6,30,New York

2025-01-04,Product B,5,15,Los Angeles 2025-01-05,Product

A,3,20,Chicago

2025-01-05,Product C,10,30,Los

Angeles import pandas as pd

Prompt the user for the file name file_name

= input()

Load the data df = pd.read_csv(file_name)

Find the month with the highest total sales df['Date'] = pd.to_datetime(df['Date'])

df['Month'] = df['Date'].dt.to_period('M').astype(str) # Convert to YYYY-MM format

df['Total Sales'] = df['Quantity'] * df['Price'] monthly_sales =

df.groupby('Month')['Total Sales'].sum()

Find the month with the highest total sales best_month

= monthly_sales.idxmax() highest_sales

= monthly_sales.max()

print(f"Best month: {best_month}") print(f"Total sales:

#{highest_sales:.2f}")

The screenshot displays a coding platform's test results interface. At the top, a summary bar shows 'Average time' as 0.082 s (82.00 ms) and 'Maximum time' as 0.182 s (182.00 ms). It also indicates that '1 out of 1 shown test case(s) passed' and '2 out of 2 hidden test case(s) passed'. Below this, 'Test case 1' is expanded, showing a duration of 182 ms and a 'Debug' button. The test case details are split into 'Expected output' and 'Actual output' columns. Both columns show the file 'sales_data.csv' and the same text output: 'Best month: 2025-01' and 'Total sales: \$1210.00'. At the bottom, there are tabs for 'Terminal' and 'Test cases', and a navigation bar with buttons for '< Prev', 'Reset', 'Submit', and 'Next >'.

4.2.2. Best Selling Product

02:08

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Find the product that sold the most in terms of quantity sold.
- Display the product that sold the most and the total quantity sold for that product.

Sample Data:

Date,Product,Quantity,Price,City

01,Product A,5,20,New York

01,Product B,3,15,Los Angeles
02,Product A,7,20,New York
02,Product C,4,30,Chicago
2025-01-03,Product B,2,15,Chicago
2025-01-03,Product A,8,20,Los Angeles 2025-01-
04,Product C,6,30,New York
2025-01-04,Product B,5,15,Los Angeles
2025-01-05,Product A,3,20,Chicago
2025-01-05,Product C,10,30,Los Angeles

Note:

The data cannot be displayed in the file. You can refer to the sample data provided for insights.

```
import pandas as pd
```

```
# Prompt the user for the file name file_name
```

```
= input()
```

```
# Load the data df = pd.read_csv(file_name) #
```

```
Find the product with the highest total
```

```
quantity sold best_product
```

```
= df.groupby('Product')['Quantity'].sum().idxmax() highest_quantity
```

```
= df.groupby('Product')['Quantity'].sum().max()
```

```
# Display the result print(f'Best selling product:
```

```
{best_product}") print(f'Total quantity sold:
```

```
{highest_quantity}")
```

2025-01-

2025-01-

2025-01-

2025-01-



4.2.3. City that Sold the Most Products

01:45

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Group the data by City and calculate the total quantity of products sold for each city.
- Find the city that sold the most products (based on the total quantity sold).

Sample Data:

Date,Product,Quantity,Price,City

2025-01-01,Product A,5,20,New York

2025-01-01,Product B,3,15,Los Angeles

2025-01-02,Product A,7,20,New York

2025-01-02,Product C,4,30,Chicago

2025-01-03,Product B,2,15,Chicago

2025-01-03,Product A,8,20,Los Angeles 04,Product

C,6,30,New York

04,Product B,5,15,Los Angeles

05,Product A,3,20,Chicago

05,Product C,10,30,Los Angeles

Note:

The data cannot be displayed in the file. You can refer to the sample data provided for insights.

```
import pandas as pd

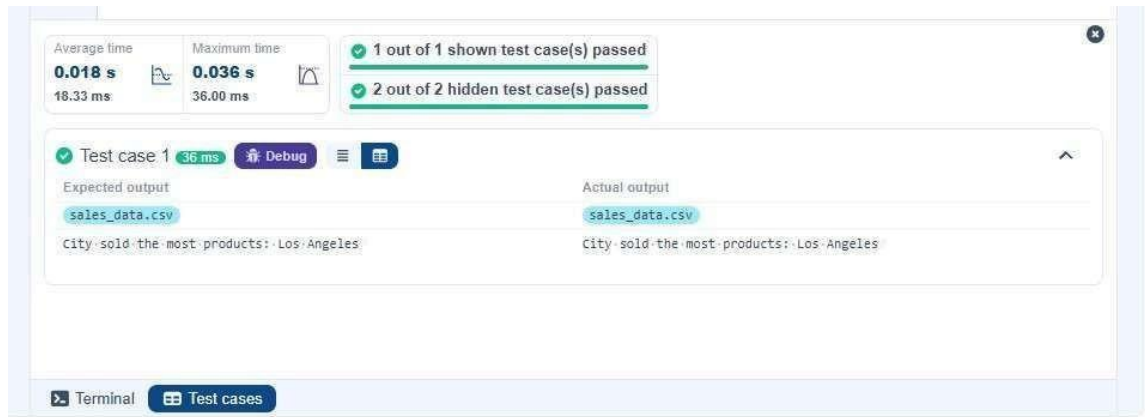
# Prompt the user for the file name file_name
file_name = input()

# Load the data df = pd.read_csv(file_name)
# Group by City and calculate total quantity city_sales =
df.groupby('City')['Quantity'].sum() # Find the city
with the highest total quantity sold best_city

best_city = city_sales.idxmax()

# Display the result print(f"City sold the most products:
{best_city}")
```

2025-01-



2025-01-

2025-01-

2025-01-

4.2.4. Most Frequently Sold Product Pairs

01:25

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the following columns: Date, Product, Quantity, Price, and City.
- For each date, find all pairs of products that were sold together (i.e., two products sold on the same date).
- Output the product pair/s that was sold most frequently.

Sample Data:

Date,Product,Quantity,Price,City

2025-01-01,Product A,5,20,New York

2025-01-01,Product B,3,15,Los Angeles

2025-01-02,Product A,7,20,New York

2025-01-02,Product C,4,30,Chicago

2025-01-03,Product B,2,15,Chicago

03,Product A,8,20,Los Angeles

04,Product C,6,30,New York

04,Product B,5,15,Los Angeles

05,Product A,3,20,Chicago 2025-01-

05,Product C,10,30,Los Angeles

Explanation:

Transactions:

- 2025-01-01: Product A, Product B
- 2025-01-02: Product A, Product C
- 2025-01-03: Product B, Product A | 2025-01-04: Product C, Product B
- 2025-01-05: Product A, Product C

Now, let's count how often the pairs of products appear together:

- Product A and Product B: Appear in transactions on 2025-01-01 and 2025-01-03.
- Product A and Product C: Appear in transactions on 2025-01-02 and 2025-01-05.
- Product B and Product C: Appears in transactions on 2025-01-04.

Most Frequent Product Combinations:

- Product A and Product B (2 times)
- Product A and Product C (2 times)

Note:

The data cannot be displayed in the file. You can refer to the sample data provided for insights.

```
import pandas as pd
from itertools import combinations
from collections import Counter
```

```
# Prompt user to input the file name file_name
```

```
= input()
```

```
# Read data from the specified CSV file df
```

```
= pd.read_csv(file_name) # write the code
```

```
pair_counter =
```

```
Counter()
```

2025-01-

Group by Date and generate product pairs

```
for _, group in df.groupby('Date'):    products
```

```
= group['Product'].tolist()    pairs =
```

```
combinations(sorted(products), 2)
```

```
pair_counter.update(pairs)
```

Find maximum frequency max_count

```
= max(pair_counter.values())
```

2025-01-

2025-01-

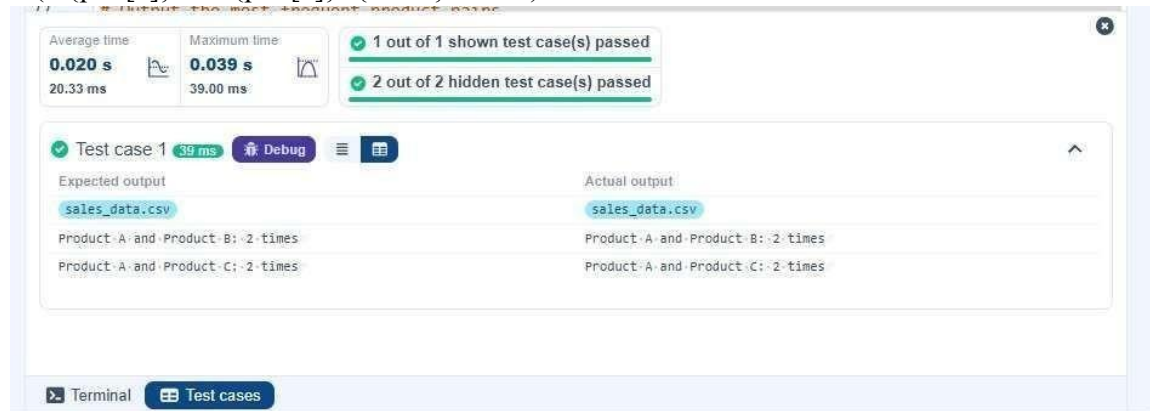
2025-01-

Output the most frequent product pairs for

```
pair, count in pair_counter.items():    if
```

```
count == max_count:
```

```
print(f" {pair[0]} and {pair[1]}: {count} times")
```



4.2.5. Titanic Dataset Analysis and Data Cleaning

01:36

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset. For each question, perform necessary data cleaning, transformations, and calculations as required.

1. Display the first 5 rows of the dataset.
2. Display the last 5 rows of the dataset.
3. Get the shape of the dataset (number of rows and columns).
4. Get a summary of the dataset (using `.info()`).
5. Get basic statistics (mean, standard deviation, etc.) of the dataset using `.describe()`.
6. Check for missing values and display the count of missing values for each column.
7. Fill missing values in the 'Age' column with the median age.
8. Fill missing values in the 'Embarked' column with the most frequent value (`mode()`).
9. Drop the 'Cabin' column due to many missing values.
10. Create a new column, 'FamilySize' by adding the 'SibSp' and 'Parch' columns.

The Titanic dataset contains columns as shown below,


```
# Load the Titanic dataset data = pd.read_csv('Titanic-Dataset.csv') # 1.
Display the first 5 rows of the dataset print(data.head())

# 2. Display the last 5 rows of the dataset print(data.tail())

# 3. Get the shape of the dataset print(data.shape)

# 4. Get a summary of the dataset (info) print(data.info())

# 5. Get basic statistics of the dataset print(data.describe())

# 6. Check for missing values print(data.isnull().sum())

# 7. Fill missing values in the 'Age' column with the median age data['Age']
= data['Age'].fillna(data['Age'].median())

# 8. Fill missing values in the 'Embarked' column with the mode data['Embarked']
= data['Embarked'].fillna(data['Embarked'].mode()[0])

# 9. Drop the 'Cabin' column due to many missing values data
= data.drop(columns=['Cabin'])

# 10. Create a new column 'FamilySize' by adding 'SibSp' and 'Parch'
data['FamilySize'] = data['SibSp'] + data['Parch']
```

Explorer

Average time: **0.166 s**
166.00 ms

Maximum time: **0.166 s**
166.00 ms

1 out of 1 shown test case(s) passed

Debugger

count	mean	std	min	25%	50%	75%	max
891.000000	446.000000	257.353842	1.000000	223.500000	446.000000	668.500000	891.000000
891.000000	0.383838	0.486592	0.000000	0.000000	0.000000	1.000000	1.000000
891.000000	2.308642	0.836071	1.000000	2.000000	3.000000	3.000000	8.000000
891.000000	0.523008	1.102743	0.000000	0.000000	0.000000	1.000000	6.000000
891.000000	0.38	0.80	0.00	0.00	0.00	0.00	6.00
1594	32.204208	6057	49.693429	0.000000	14.454208	31.000000	512.329200

[8 rows x 7 columns]

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	0	0	0	0	177	0	0	0	0	687	2

dtype: int64

Terminal Test cases

< Prev Reset Submit Next >

9:37 PM
4/24/2026

4.2.6. Titanic Dataset Analysis and Data Cleaning - 2

02:18

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Create a new column 'IsAlone' which is 1 if the passenger is alone (FamilySize = 0), otherwise 0.
2. Convert the 'Sex' column to numeric values (male: 0, female: 1).
3. One-hot encode the 'Embarked' column, dropping the first category.
4. Get the mean age of passengers.
5. Get the median fare of passengers.
6. Get the number of passengers by class.
7. Get the number of passengers by gender.
8. Get the number of passengers by survival status.
9. Calculate the survival rate of passengers.
10. Calculate the survival rate by gender.

The Titanic dataset contains columns as shown below,

Sample Data:

PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked

1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S

2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC	17599,71.2833,C85,C
--	---------------------

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarke
-------------	----------	--------	------	-----	-----	-------	-------	--------	------	-------	---------

3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S

4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S

5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S

6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q

7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S

8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S

9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S

10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

```
import pandas as pd
import numpy as np
```

```
# Load the Titanic dataset data =
pd.read_csv('Titanic-Dataset.csv')

data['FamilySize'] = data['SibSp'] +
data['Parch']

# 1. Create a new column 'IsAlone' (1 if alone, 0 otherwise) data['IsAlone']
= np.where(data['FamilySize'] == 0, 1, 0)
```

2. Convert 'Sex' to numeric (male: 0, female: 1) data['Sex']

= data['Sex'].map({'male': 0, 'female': 1})

3. One-hot encode the 'Embarked' column data = pd.get_dummies(data, columns=['Embarked'],
drop_first=True) # 4.

Get the mean age of passengers print(data['Age'].mean())

5. Get the median fare of passengers print(data['Fare'].median())

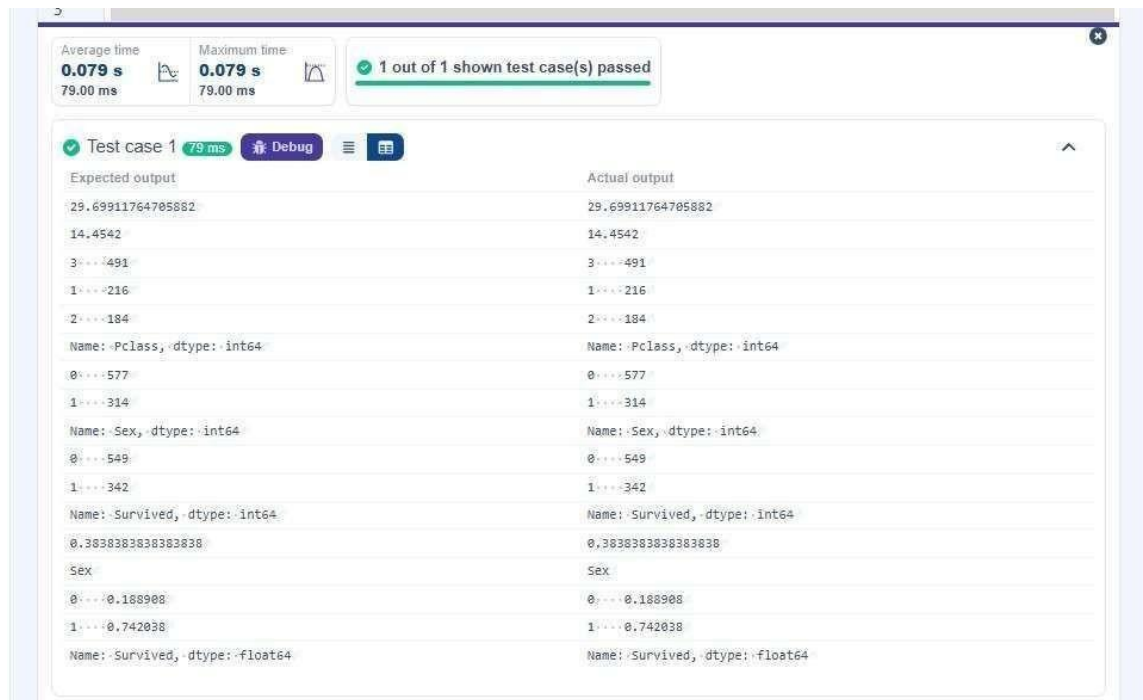
6. Get the number of passengers by class print(data['Pclass'].value_counts())

7. Get the number of passengers by gender print(data['Sex'].value_counts())

8. Get the number of passengers by survival status print(data['Survived'].value_counts())

9. Calculate the survival rate print(data['Survived'].mean())

10. Calculate the survival rate by gender print(data.groupby('Sex')['Survived'].mean())



4.2.7. Titanic Dataset Analysis and Data Cleaning - 3

03:10

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Calculate the survival rate by class.
2. Calculate the survival rate by embarkation location (Embarked_S).
3. Calculate the survival rate by family size (FamilySize).
4. Calculate the survival rate by being alone (IsAlone).
5. Get the average fare by passenger class (Pclass).

8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S

9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S

10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

Note: Refer to the visible test case for better reference.

```
import pandas as pd
import numpy
```

```
as np
```

```
# Load the Titanic dataset data = pd.read_csv('Titanic-Dataset.csv')
```

```
data['FamilySize'] = data['SibSp'] + data['Parch'] data['IsAlone'] =
```

```
np.where(data['FamilySize'] > 0, 0, 1) data = pd.get_dummies(data,
```

```
columns=['Embarked'], drop_first=True) # 1. Calculate the survival rate
```

```
by class print(data.groupby('Pclass')['Survived'].mean())
```

```
# 2. Calculate the survival rate by embarked location
```

```
print(data.groupby('Embarked_S')['Survived'].mean()) # 3. Calculate the survival rate by family size
```

```
print(data.groupby('FamilySize')['Survived'].mean())
```

```
# 4. Calculate the survival rate by being alone print(data.groupby('IsAlone')['Survived'].mean()) #
```

```
5. Get the average fare by class print(data.groupby('Pclass')['Fare'].mean())
```

```
# 6. Get the average age by class print(data.groupby('Pclass')['Age'].mean())
```

```
# 7. Get the average age by survival status print(data.groupby('Survived')['Age'].mean())
```

```
# 8. Get the average fare by survival status print(data.groupby('Survived')['Fare'].mean())
```


9. Get the number of survivors by class print(data[data['Survived'] == 1]['Pclass'].value_counts())

10. Get the number of non-survivors by class print(data[data['Survived'] == 0]['Pclass'].value_counts())

